# 15.438 Fixed Income: HW4

Trisha Razdan, trazdan@mit.edu
Corentin Venet, venetco@mit.edu

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### 1.1 a

**Three-Period BDT Binomial Tree (σ = 14%, Given Spot Yields: 3.50%, 4.01%, 4.531%)**

**INPUTS**

| Period | Spot Yield | Price per 100 | 2×σ (Volatility Input) |
|---|---|---|---|
| 1 | 3.500% | 96.618357 | 0.28 |
| 2 | 4.010% | 92.437844 | 0.28 |
| 3 | 4.531% | 87.551720 | 0.28 |

**BINOMIAL TREE OF SHORT RATES**

| State \ Period | t = 1 | t = 2 | t = 3 |
|---|---|---|---|
| High (uu) | 3.5000% | 5.1559% | 7.2624% |
| Mid (ud) | - | 3.8967% | 5.4888% |
| Low (dd) | - | - | 4.1483% |

**NOTES**

σ = 14% → volatility ratio: r_u / r_d = e^(2σ) = e^0.28 ≈ 1.3231
r(t=1) = 3.50% [single node, derived directly from P1]
r_d(t=2) = 3.8967%, r_u(t=2) = 5.1559% [calibrated to P2]
r_dd(t=3) = 4.1483%, r_ud(t=3) = 5.4888%, r_uu(t=3) = 7.2624% [calibrated to P3]
All rates satisfy: r_u = r_d × e^(2σ) at each period (BDT structure)

### b. Callable Bond at \$99.50, $\sigma = 14\%$

Using the $\sigma = 14\%$ BDT tree, we back-induct through the tree applying the American call constraint ex-coupon at $t = 1, 2$.

**At $t = 2$:** The ex-coupon continuation values are:

$$V_u = \frac{105.25}{1.051559} = 100.089, \qquad V_d = \frac{105.25}{1.038967} = 101.303$$

Both exceed \$99.50, so the bond is called at both nodes: $V_u = V_d = 99.50$.

**At $t = 1$:**

$$V_0^{ex-coup} = \frac{0.5(99.50) + 0.5(99.50)}{1.035} = 96.135 \quad < 99.50 \Rightarrow notcalled$$

$$P_{callable} = 96.135 + 5.25 = \$101.385$$

The straight bond price is $P_{nc} = 102.364$, giving an **option value of \$0.979**. The yield spread between callable and non-callable is **larger** than at lower volatility: higher $\sigma$ fattens the tails of the rate distribution, increasing the probability the call lands in-the-money and making the embedded option more valuable to the issuer.

## c. Adding an American Put at \$99.00

At each node we apply $V^* = \max(99.00,\ \min(V^{nc},\ 99.50))$.

At $t = 2$ both nodes are called to 99.50 (same as above; neither triggers the put floor). Rolling back to $t = 1$:

$$V^{ex-coup} = \frac{99.50}{1.035} = 96.135 \quad \Rightarrow \quad \max(99.00,\ \min(96.135, 99.50)) = 99.00 \quad \textbf{(put exercised)}$$

$$P_{call+put} = \frac{99.00 + 5.25}{1.035} = \$101.690$$

**Put option value** $= 101.690 - 101.385 = \$0.305$.

The **optimal put strategy** is to exercise at $t = 1$: the discounted continuation value (96.14) falls below the \$99 floor, so the holder puts the bond back to the issuer. At $t = 2$ the call dominates and the put is never reached.

## d. Declining Call Price Schedule (\$101.10 at $t = 1$, \$100.50 at $t = 2$)

At $t = 2$, applying the \$100.50 call price:

$$V_u = 100.089 < 100.50 \Rightarrow notcalled, \qquad V_d = 101.303 > 100.50 \Rightarrow calledto100.50$$

Rolling back to $t = 1$ with the \$101.10 strike:

$$V^{ex-coup} = \frac{0.5(100.089) + 0.5(100.50)}{1.035} = \frac{100.295}{1.035} = 96.900 \quad < 101.10 \Rightarrow notcalled$$

$$P_{callable} = 96.900 + 5.25 = \$102.150$$

**Option value** $= 102.364 - 102.150 = \$0.214$, well below the \$0.979 from part (b). The high early call price (\$101.10) provides meaningful call protection at $t = 1$, and only the down-node at $t = 2$ is called. With fewer nodes exercised, the embedded option is worth far less to the issuer and the bond trades closer to its straight value.

# 2 2

For the entire problem, we used the closed-form CIR zero-coupon bond price under the risk-neutral measure $dr = \beta(\bar{r} - r)\,dt + \sigma\sqrt{r}\,dW$:

$$P(0,\tau) = A(\tau)\,e^{-B(\tau)\,r_0}, \qquad \gamma = \sqrt{\beta^2 + 2\sigma^2},$$

$$B(\tau) = \frac{2(e^{\gamma\tau} - 1)}{(\gamma+\beta)(e^{\gamma\tau} - 1) + 2\gamma}, \qquad A(\tau) = \Big[2\gamma\, e^{(\beta+\gamma)\tau/2}(\gamma+\beta)(e^{\gamma\tau} - 1) + 2\gamma\Big]^{2\beta\bar{r}/\sigma^2}.$$

Continuous-compound yields $y_c(\tau) = -\ln P(0,\tau)/\tau$ are converted to a bond-equivalent (semi-annual) basis via $y_{beb} = 2(e^{y_c/2} - 1)$, since the target yields are quoted b.e.b.

As suggested, we wrote our own version of code to implement the closed-from CIR on a monthly basis in Python.

## a. CIR calibration

In order to find the best parameters for our CIR model, we deciced to calibrate it through an automated parameter search in Python

First, we found out through train and error that the shape of the term structure is largely dominated by $(r_0, \beta, \bar{r})$. This can be explain theoretically by the fact that $\sigma$ enters only through a small convexity correction and is therefore not pinned down by curve fit alone. Therefore, we decided to fix $\sigma = 0.08$ (consistent with the spreadsheet default) and choose $(r_0, \beta, \bar{r})$ by minimizing the sum of squared errors at the four focus points $\tau \in \{1, 5, 10, 30\}$. Here are the results that we got:

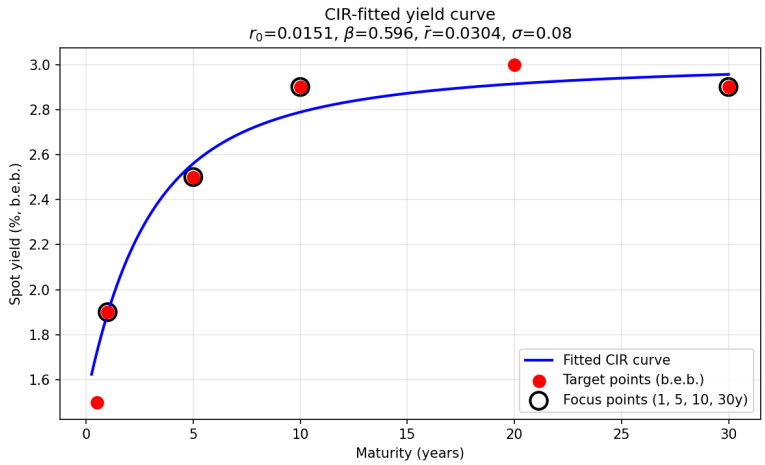
$$r_0 = 0.01509, \qquad \beta = 0.5961, \qquad \bar{r} = 0.03044, \qquad \sigma = 0.08$$

In addition, the Feller condition $2\beta\bar{r} = 0.0363 > \sigma^2 = 0.0064$ is comfortably satisfied, so the simulated short rate stays strictly positive.

With the parameters found above, here are the predictions with our model, compared to the target rates :

| $\tau$ (yr) | Target (b.e.b.) | Model (b.e.b.) | Error (bp) |
|---|---|---|---|
| 0.5 | 1.500% | 1.723% | +22.3 |
| 1.0 | 1.900% | 1.895% | −0.5 |
| 5.0 | 2.500% | 2.560% | +6.0 |
| 10.0 | 2.900% | 2.789% | −11.2 |
| 20.0 | 3.000% | 2.914% | −8.6 |
| 30.0 | 2.900% | 2.956% | +5.6 |

The sum of squared errors over the four focus points is SSE $= 1.92 \times 10^{-6}$, i.e. MSE $= 4.81 \times 10^{-7}$. The 1y point is essentially exact and the 5/10/30y points are within ∼11 bp; the curve overprices the 6m point by 22 bp, which is the price paid for a smooth single-factor fit through a steeply rising front end.

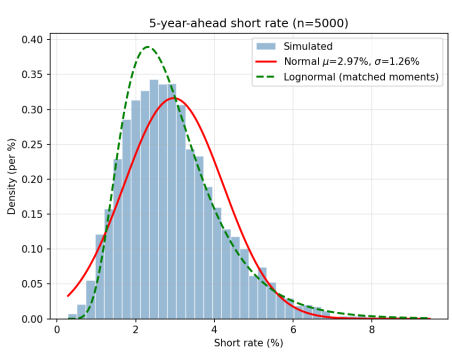


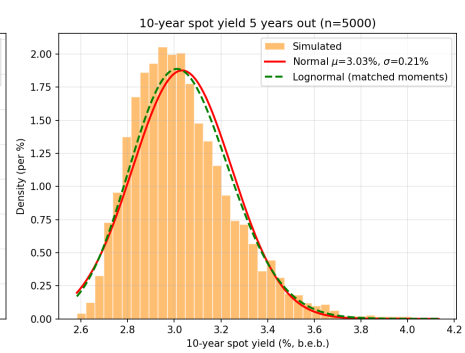

## b. Histograms five years out

Using the calibration above we generate $N = 5{,}000$ Monte-Carlo paths of the short rate on a monthly grid ($\Delta t = 1/12$) over 60 months. From each terminal short rate $r(5)$ we then price a 10-year zero using the closed-form CIR formula and convert to b.e.b. yield. Here are the empirical moments obtained :

| | 5y-ahead $r$ | 10y yield 5y out (b.e.b.) |
|---|---|---|
| mean | 2.97% | 3.03% |
| std | 1.26% | 0.21% |
| skewness | +0.86 | +0.86 |
| excess kurtosis | +1.07 | +1.08 |

The 10y yield is far less dispersed than $r(5)$ because the 10-year bond formula averages the short rate over a long horizon and the term-structure mapping $r \mapsto y_{10}(r)$ is nearly linear and heavily compressed by mean reversion.



## c. Normal vs. lognormal

Both empirical distributions are right-skewed (skewness $\approx 0.86$).

For the **short rate**, the **lognormal density** (matched to the simulated mean and variance) **is visibly the better fit**: it captures the hard non-negativity floor and the long right tail, whereas the symmetric Normal places non-trivial mass below zero and underweights the upper tail.

However, because $y_{10}$ is an integral-like average over a long horizon, the dispersion is small (s.d. of only 21 bp) and the support sits well away from zero, so the asymmetry of the lognormal matters much less. Visually, Normal and Lognormal are near-indistinguishable for the **10y yield**, **either is a reasonable summary**.

## d. European call on a 1.5-year T-Note

Payoff description : The call expires at $T_{opt} = 2$ years. The deliverable is then a just-issued 1.5-year T-Note with semi-annual 2.5% coupon and \$100,000 face; strike is $K =$\$100,000 (par).

At delivery, the bond pays \$1,250 at $t = 0.5, 1.0$ and \$101,250 at $t = 1.5$ from delivery, so its price at time $T$ is (as a function of the short rate $r(T)$) :

$$B(T; r) = 1250\, P^{CIR}(r, 0.5) + 1250\, P^{CIR}(r, 1.0) + 101250\, P^{CIR}(r, 1.5).$$

Path-wise we accumulate the discount factor $D = \exp\big(-\int_0^T r(s)\, ds\big)$ , and the call price is

$$C = E^Q\big[D \cdot \max(B(T; r(T)) - K, 0)\big].$$

With 50,000 Monte-Carlo paths (monthly steps), we get:

$$C \;=\; \$228.68 \;\; (s.e.\ \$1.67,\ 95\%CI[\$225.40,\ \$231.95])$$

Additionally, about 40.3% of paths finish in-the-money.

Two consistency checks confirm the simulation is correctly risk-neutral: the MC mean discount factor matches the closed-form P(0,2) (0.95785 vs. 0.95804) and the MC-implied forward bond price matches the closed-form forward (\$99,634 vs. \$99,648), both within MC standard error. These checks validate the discretization used in the Monte Carlo simulation, since the option payoff is non-linear and has no closed form to compare against directly.

## e. Extra credit

**One-factor fit.** Repeating the procedure of part (a) on the new target curve with $\sigma = 0.08$ yields

$$r_0 = 0.05757, \qquad \beta = 0.7457, \qquad \bar{r} = 0.04513, \qquad \text{MSE} = 4.06 \times 10^{-6}.$$

The single-factor model produces a strictly monotone curve, so it cannot reproduce the fact that the curve is inverted at the very front end (the 1y point at 5.8% sits *above* the 6m point at 5.4%) nor the slight 20-year hump.

**Two-factor extension.** We model the short rate as the sum of two independent CIR factors, $r(t) = r_1(t) + r_2(t)$, each with its own $(\beta_i, \bar{r}_i, \sigma_i)$. Under independence the zero-coupon bond price factorizes as $P(0,\tau) = P_1(0,\tau)P_2(0,\tau)$, so the spot-yield curve is the sum of the two component yield curves.

We calibrate all eight parameters by differential evolution. Indeed, this method allows large jumps and prevents the optimization to be trapped into a local minimum the way basic gradient-descent methods do, especially based on the fact that the loss function is non-convex here. The 8-parameter optimization gives :

| | $r_{i,0}$ | $\beta_i$ | $\bar{r}_i$ | $\sigma_i$ |
|---|---|---|---|---|
| Factor 1 | 0.04736 | 0.3379 | 0.01001 | 0.1000 |
| Factor 2 | 0.01001 | 0.1469 | 0.04536 | 0.0998 |

The economic intuition behind theses 2 factors can be interpreted as following: factor 1 generates the front-end inversion (starts high and reverts down) while factor 2 lifts the long end (starts low and reverts up).

Here are the prediction results :

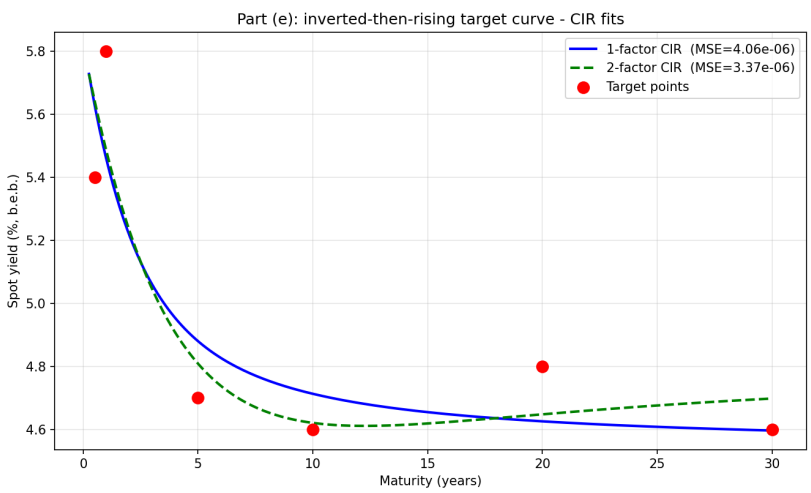
| $\tau$ | Target | 1-fac | 2-fac | 1f err (bp) | 2f err (bp) |
|---|---|---|---|---|---|
| 0.5 | 5.400% | 5.628% | 5.641% | +22.8 | +24.1 |
| 1.0 | 5.800% | 5.459% | 5.486% | −34.1 | −31.5 |
| 5.0 | 4.700% | 4.880% | 4.808% | +18.0 | +10.8 |
| 10.0 | 4.600% | 4.713% | 4.620% | +11.3 | +2.0 |
| 20.0 | 4.800% | 4.625% | 4.648% | −17.5 | −15.3 |
| 30.0 | 4.600% | 4.596% | 4.698% | −0.4 | +9.7 |
| MSE | | | | $4.06 \times 10^{-6}$ | $3.37 \times 10^{-6}$ |

Compared to the 1-factor MSE, the 2-factor MSE is **17% lower**, and the gain is concentrated where the 1-factor model fails: the 5y and 10y points improve from +18 and +11 bp to +11 and +2 bp respectively, because the curvature created by superposing one falling and one rising factor matches the data far better than any monotone shape.

**Monte Carlo verification.** To verify the analytical curve, we simulate both factors on a monthly grid over 10,000 paths, compute the discount factor $D = \exp\left(-\int_0^\tau (r_1 + r_2)\, ds\right)$ on each path, and estimate $P(0,\tau) \approx E^Q[D]$ by averaging. Converting to b.e.b. yields gives:

| $\tau$ | Closed form | MC (10,000 paths) |
|---|---|---|
| 0.5 | 5.641% | 5.623% |
| 1.0 | 5.486% | 5.462% |
| 5.0 | 4.808% | 4.778% |
| 10.0 | 4.620% | 4.608% |

The comparison of the results between the closed form and the Monte Carlo shows that the difference for every $\tau$ is within $\sim$3 bp, well inside Monte-Carlo standard error, thus confirming the analytical curve obtained with the 2-factor model.




There remains a clear residual misfit at the 1y and 20y points. We believe that this reflects a genuine non-monotonic "smile" in the target that no two-factor affine model with independent CIR factors can produce exactly. Perhaps another model that allows some correlation between the Brownian motions, or going to a three-factor model (level, slope, curvature), would close the remaining gap.